Classification of Traffic Accident Severity Using the K-Nearest Neighbor (KNN) Algorithm

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***Abstract*—** Traffic accidents remain a major public safety problem in many countries, including Indonesia, causing material losses, injuries, and fatalities each year. This study applies the K-Nearest Neighbor (KNN) algorithm to classify the severity of traffic accidents into three categories: Fatal, Severe Injury, and Minor Injury, based on ten predictor variables covering victim demographics, environmental conditions, and accident circumstances. The dataset consists of 300 traffic accident records collected with attributes including victim age, gender, time of occurrence, vehicle type, weather condition, road type, use of safety equipment, vehicle speed, and the number of vehicles involved. Categorical features were transformed using label encoding and numerical features were standardized before model training. The dataset was split into 80% training data and 20% testing data. A KNN classifier with k = 5 and Euclidean distance achieved an accuracy of 70% on a stratified test split, with the highest per-class performance obtained for the Fatal category (F1-score = 0.82). To evaluate the stability of the model, 200 independent train-test splits were performed, yielding a mean accuracy of 54.35% (SD = 6.09%), while a sensitivity analysis over k = 1 to 20 indicated that k = 7 produced the best average performance (76.67%). These results suggest that KNN can capture meaningful patterns between accident characteristics and injury severity, although classification performance is sensitive to the choice of k and to variability across random data partitions. The proposed approach can support transportation authorities in prioritizing safety interventions based on accident risk factors.

***Keywords*—** *K-Nearest Neighbor; traffic accident; classification; severity; machine learning*

I. INTRODUCTION

Traffic accidents continue to be one of the most pressing public safety issues worldwide, and Indonesia is no exception. The high number of accidents recorded every year not only results in material losses but also causes fatalities and injuries with varying levels of severity. Several factors are known to influence the severity of a traffic accident, including the victim's age and gender, the time of occurrence, the type of vehicle involved, weather conditions, road type, the use of safety equipment, vehicle speed, and the number of vehicles involved in the incident.

With the advancement of information technology, the application of machine learning has become an effective approach for analyzing patterns in traffic accident data. Machine learning enables the discovery of relationships between variables based on historical data, allowing the classification or prediction of future occurrences. By leveraging available data, a trained model can help identify the severity level of an accident faster and more objectively than manual analysis.

In this study, the K-Nearest Neighbor (KNN) algorithm is used as the classification method. KNN was selected because of its conceptual simplicity, ease of implementation, and its ability to perform well in classifying data based on the similarity between data points. KNN determines the class of a new data point based on the majority class among its k nearest neighbors, computed using a distance metric, making it well suited for the accident-severity classification task addressed in this study.

The remainder of this paper is organized as follows. Section II reviews related work on machine learning approaches for traffic accident severity analysis. Section III describes the dataset and the proposed KNN-based research method. Section IV presents the experimental results and discussion. Finally, Section V concludes the paper and outlines directions for future work.

II. RELATED WORK

The K-Nearest Neighbor algorithm is a non-parametric, instance-based classification method that assigns a class label to a query point based on the majority vote of its k closest neighbors in the feature space, as originally formulated by Fix and Hodges [1] and later analyzed by Cover and Hart [2]. Because it makes no assumption about the underlying data distribution, KNN is frequently used as a baseline classifier for tabular, mixed-type datasets such as traffic accident records.

Prior studies in the transportation safety domain have shown that supervised classification algorithms, including KNN, decision trees, and ensemble methods, can be used to relate accident circumstances—such as road geometry, weather, and driver behavior—to the resulting injury severity, supporting evidence-based road-safety planning. Building on this line of research, the present study focuses specifically on the application of KNN, following the standard scikit-learn implementation described by Pedregosa et al. [3], to a locally collected traffic accident dataset consisting of ten predictor variables and a three-class severity target.

III. RESEARCH METHOD

A. Dataset

The dataset used in this study consists of 300 traffic accident records with 10 predictor attributes and 1 target attribute, as summarized in Table I. The target variable, Severity Category, has three classes: Fatal (125 records, 41.7%), Severe Injury (115 records, 38.3%), and Minor Injury (60 records, 20.0%), indicating a moderately imbalanced class distribution.

TABLE 1
DATASET FEATURE DESCRIPTION

Feature	Description	Type
Victim_Age	Age of the victim (years)	Numeric
Gender	Male / Female	Categorical
Time_of_Occurrence	Hour of the incident (0-23)	Numeric
Time_Period	Morning/Afternoon/Evening/Night	Categorical
Vehicle_Type	Motorcycle, Car, Truck, Bus, Bicycle	Categorical
Weather_Condition	Clear, Rain, Fog	Categorical
Road_Type	Urban, Rural, Intercity, Toll road	Categorical
Safety_Equipment	Use of safety equipment (Yes/No)	Categorical
Speed_Kmh	Vehicle speed (km/h)	Numeric
Vehicles_Involved	Number of vehicles involved	Numeric
Severity_Category	Fatal / Severe Injury / Minor Injury	Target

B. Data Preprocessing

Prior to modelling, the identifier column (No) was removed since it carries no predictive information. The six categorical attributes (Gender, Time Period, Vehicle Type, Weather Condition, Road Type, and Safety Equipment) were transformed into numeric form using label encoding. The four numeric attributes were then standardized using z-score normalization (StandardScaler), so that all features contribute proportionally to the Euclidean distance computation used by KNN. The dataset was subsequently split into 80% training data (240 records) and 20% testing data (60 records).

C. KNN Classification

The K-Nearest Neighbor classifier was implemented using scikit-learn's KNeighborsClassifier. Given a query instance, the algorithm computes the Euclidean distance to all training instances, selects the k nearest neighbors, and assigns the majority class among them, with uniform weighting for all neighbors. The configuration of the final model is summarized in Table II.

TABLE 2
KNN MODEL CONFIGURATION

Component	Value
Algorithm	K-Nearest Neighbor
Number of Neighbors (k)	5
Distance Metric	Minkowski (p=2, Euclidean)
Weight Function	Uniform
Number of Features	10
Training Data	240 records
Testing Data	60 records

D. Evaluation Procedure

Model performance was assessed using accuracy, precision, recall, and F1-score derived from the confusion matrix. Because a single train-test split can be sensitive to the particular random partition, the experiment was repeated over 200 independent random splits (random_state = 1 to 200) using k = 5, from which the mean and standard deviation of accuracy were computed to assess model stability. In addition, a sensitivity analysis was performed by varying k from 1 to 20 on a stratified 80:20 split to identify the value of k that yields the best classification performance.

IV. RESULTS AND DISCUSSION

A. Stability Across 200 Random Splits

Fig. 1 shows the accuracy obtained across 200 independent train-test splits with k = 5. The mean accuracy across all trials was 54.35% (SD = 6.09%), with a maximum of 70.00% and a minimum of 38.33%. This relatively wide range indicates that model performance is sensitive to how the data is partitioned, which is expected given the moderate dataset size (300 records) relative to the number of classes and features.

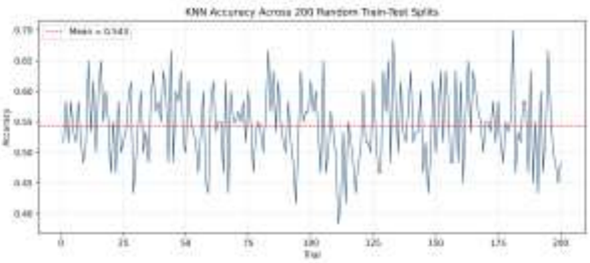

Fig. 1. KNN accuracy across 200 random train-test splits (k = 5).

B. Effect of K on Accuracy

The sensitivity analysis over k = 1 to 20, illustrated in Fig. 2, shows that accuracy is relatively low for very small values of k (e.g., k = 1 gives 56.67%, k = 2 gives 48.33%) due to sensitivity to noise, then improves as k increases. The highest accuracy, 76.67%, was obtained at k = 7, followed by k = 6 (73.33%) and k = 8 (71.67%). Beyond k = 8, accuracy gradually declines, suggesting that overly large neighborhoods begin to blur the boundaries between severity classes. Although the original model configuration used k = 5 (accuracy = 70.00%), these results indicate that k = 7 would be a marginally better choice for this dataset.

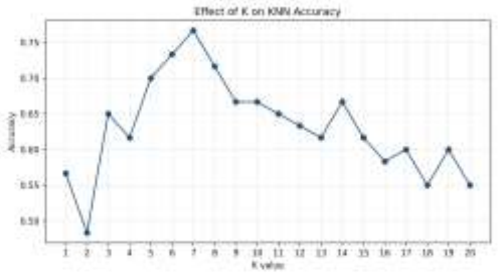

Fig. 2. Effect of the number of neighbors (k) on classification accuracy.

C. Final Model Performance (k = 5)

Using a stratified 80:20 split, the final KNN model with k = 5 achieved an overall accuracy of 70.00% on the test set. Table III presents the precision, recall, and F1-score for each severity class, and Table IV presents the corresponding confusion matrix, visualized in Fig. 3.

TABLE 3
CLASSIFICATION REPORT (k = 5)

Class	Precision	Recall	F1-Score	Support
Severe Injury	0.67	0.61	0.64	23
Minor Injury	0.53	0.67	0.59	12
Fatal	0.83	0.80	0.82	25
Accuracy			0.70	60
Macro avg	0.68	0.69	0.68	60
Weighted avg	0.71	0.70	0.70	60

TABLE 4
CONFUSION MATRIX (k = 5)

Actual \ Predicted	Severe	Minor	Fatal
Severe Injury	14	6	3
Minor Injury	3	8	1
Fatal	4	1	20

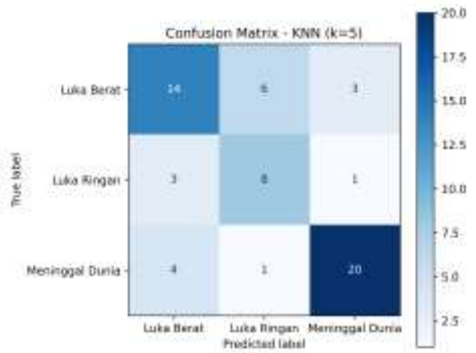

Fig. 3. Confusion matrix of the KNN model (k = 5) on the test set.

D. Discussion

The results show that KNN performs best in identifying Fatal cases (F1-score = 0.82), suggesting that fatal accidents in this dataset are associated with more distinctive feature patterns—likely reflecting the combined influence of speed, vehicle type, and lack of safety-equipment use. In contrast, the Minor Injury class shows lower precision (0.53), and the confusion matrix indicates that Severe Injury cases are the most frequently misclassified, primarily confused with Minor Injury and Fatal outcomes. This overlap is consistent with the inherently continuous nature of injury severity, where the boundary between adjacent severity levels is not always sharply defined by the recorded attributes.

V. CONCLUSION

This study applied the K-Nearest Neighbor algorithm to classify the severity of traffic accidents into three categories using a dataset of 300 records and ten predictor attributes. The final model, using k = 5 and Euclidean distance, achieved a test accuracy of 70.00%, with the strongest performance observed for the Fatal class (F1-score = 0.82). A stability analysis over 200 random splits showed a mean accuracy of 54.35% (SD = 6.09%), while a sensitivity analysis indicated that k = 7 provides a marginally higher average accuracy (76.67%) than the originally selected k = 5. These findings demonstrate that KNN is a viable baseline method for accident-severity classification and can provide useful, data-driven insight to support road-safety decision-making. Future work may explore additional features, alternative distance metrics, class-imbalance handling, and comparison with other classification algorithms such as Random Forest or Support Vector Machines to further improve predictive performance.

REFERENCES

[1] E. Fix and J. L. Hodges, "Discriminatory Analysis. Nonparametric Discrimination: Consistency Properties," USAF School of Aviation Medicine, Randolph Field, Texas, Report No. 4, 1951.

[2] T. M. Cover and P. E. Hart, "Nearest neighbor pattern classification," IEEE Transactions on Information Theory, vol. 13, no. 1, pp. 21-27, 1967.

[3] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825-2830, 2011.

[4] World Health Organization, "Global status report on road safety 2023," Geneva, Switzerland, 2023.

[5] S. Raschka, Y. H. Liu, and V. Mirjalili, Machine Learning with PyTorch and Scikit-Learn, Packt Publishing, 2022.

Note: Domain-specific literature references relevant to the author's own review of prior traffic-accident-severity studies should be added by the author from verified sources before submission.